

Derivatives

Recall the Rates of Change

We need to look at the rate of change problem. Here we are going to consider a function $f(x)$ that represents some quantity that varies as x varies.

what we want to do here is determine just how fast $f(x)$ is changing at some point from $x=x_1$ to $x=x_0$ over the interval $[x_0, x_1]$. This is called average rate of change of $f(x)$ at $x=x_0$ over the interval $[x_0, x_1]$ and we write

$$r_{\text{ave}} = \frac{f(x_1) - f(x_0)}{x_1 - x_0} = \frac{\text{Change in } f(x)}{\text{Change in } x}$$

To estimate the instantaneous rate of change at $x=x_0$, we need to do is to choose values of x getting closer and closer to x_0 .

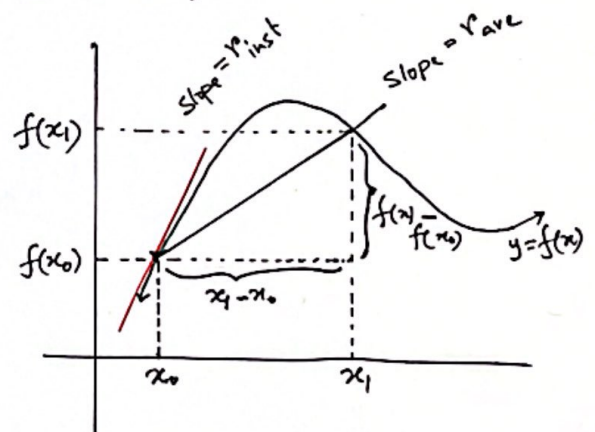
For instantaneous rate of change we can write,

$$r_{\text{inst}} = \lim_{x_1 \rightarrow x_0} \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

let $h = x_1 - x_0$, since $x_1 \rightarrow x_0$

so $h \rightarrow 0$ ~~which is~~

$$r_{\text{inst}} = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}$$



If this limit exists, then it can be interpreted as the slope of the tangent line to the curve $y = f(x)$ at $x = x_0$.

This limit is so important that it has a special notation:

$$f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

Theorem: If f is differentiable at a , then f is continuous at a .

Proof: We want to prove that f is continuous at a i.e.

$$\lim_{x \rightarrow a} f(x) = f(a).$$

By the definition of differentiability at a ,

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$$\text{Now, } f(x) - f(a) = \frac{f(x) - f(a)}{x - a} \cdot (x - a)$$

$$\begin{aligned} \lim_{x \rightarrow a} [f(x) - f(a)] &= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} (x - a) \\ &= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \cdot \lim_{x \rightarrow a} (x - a) \\ &= f'(a) \cdot \lim_{x \rightarrow a} (x - a) \\ &= 0 \end{aligned}$$

$$\text{Thus } \lim_{x \rightarrow a} f(x) = f(a).$$

Therefore, f is continuous at a .

□

Problem: Show that

$$f(x) = \begin{cases} x^2 + 1, & x \leq 1 \\ 2x, & x > 1 \end{cases}$$

is continuous and differentiable at $x=1$.

Solⁿ: If the function is differentiable at $x=1$, then it must be continuous at $x=1$.

Now,

$$f'_-(x) = \lim_{h \rightarrow 0^-} \frac{f(x+h) - f(x)}{h}$$

$$f'_-(1) = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2 + 2h + h^2 - 2}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(2+h)}{h} = 2$$

and,

$$f'_+(x) = \lim_{h \rightarrow 0^+} \frac{f(x+h) - f(x)}{h}$$

$$f'_+(1) = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h}$$

$$\lim_{h \rightarrow 0} \frac{2 + 2h - 2}{h}$$

$$\lim_{h \rightarrow 0} \frac{2h}{h} = 2$$

$$\cancel{\text{and}} \quad f'_-(1) = f'_+(1)$$

Thus

Therefore $f(x)$ is differentiable at $x=1$.

✱

$$\begin{aligned} f(x) &= x^2 + 1 \\ f(1) &= 1^2 + 1 = 2 \end{aligned}$$

$$\begin{aligned} f(1+h) &= (1+h)^2 + 1 \\ &= 1 + 2h + h^2 + 1 \\ &= 2 + 2h + h^2 \end{aligned}$$

$$f(x) = 2x$$

$$f(1) = 2$$

$$\begin{aligned} f(1+h) &= 2(1+h) \\ &= 2 + 2h \end{aligned}$$

Problem: show that $f(x) = \begin{cases} x^2+2 & x \leq 1 \\ x+2 & x > 1 \end{cases}$

is continuous but not differentiable at $x=1$.

Do yourself!!!

Problem: Find the derivative with respect to x of $f(x) = x^3 - x$.

Solⁿ:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(x+h)^3 - (x+h) - [x^3 - x]}{h}$$

$$= \lim_{h \rightarrow 0} \frac{x^3 + 3x^2h + 3xh^2 + h^3 - x - h - x^3 + x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{3x^2h + 3xh^2 + h^3 - h}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(3x^2 + 3xh + h^2 - 1)}{h}$$

$$= \lim_{h \rightarrow 0} (3x^2 + 3xh + h^2 - 1)$$

$$= 3x^2 - 1$$

□